VIX Tupiniquim III: Dynamic Measurement of Financial Stress via TVP-VAR-SV

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Abstract

This paper develops the VIX Tupiniquim III, a domestic financial stress indicator built upon a Time-Varying Parameter Vector Autoregressive model with Stochastic Volatility (TVP-VAR-SV). The system combines four representative variables of the Brazilian financial market: sovereign risk, measured by the 5-year sovereign Credit Default Swap (CDS 5Y); the benchmark monetary policy rate, represented by Selic Over; the equity market, measured by the monthly return of the Ibovespa index; and domestic credit conditions, represented by the retail banking credit spread. The primary feature of the model is allowing both the shock transmission mechanism and the system's volatility to evolve over time. Model parameters are treated as latent state variables whose trajectories are inferred from observable data. Estimation is carried out in a Bayesian framework combining state-space filtering algorithms and Gibbs Sampling. From the estimated conditional covariance matrix, a synthetic financial stress index is derived based on its maximum eigenvalue and subsequently normalized to a historical baseline mean of 100. The indicator's dynamic trajectory is evaluated against official Brazilian economic recessions dated by CODACE/FGV and tested for temporal precedence against the Economic Policy Uncertainty Index (Brazil EPU). The empirical findings show that the indicator effectively tracks historical episodes of macro-financial distress, distinguishing between persistent structural deteriorations and abrupt liquidity shocks. Temporal precedence tests indicate no linear predictive relationship between the VIX Tupiniquim III and Brazil EPU, confirming that market-implied and news-based uncertainty capture distinct dimensions of economic distress.

Keywords: Financial Stress; TVP-VAR; Stochastic Volatility; Bayesian Econometrics; Temporal Precedence; VIX Tupiniquim.

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1 Introduction

Financial stress is inherently dynamic. How distinct market segments absorb and transmit shocks changes across business cycles, as does the underlying magnitude of shocks hitting the economic system.

During tranquil periods, for instance, an increase in sovereign risk may generate relatively muted spillovers to other domestic markets. In periods of macro-financial fragility, however, the exact same shock can propagate with significantly higher intensity across equity valuations, monetary policy stances, and banking credit spreads.

This characteristic highlights a critical limitation of traditional constant-parameter models. Although conventional Vector Autoregressions (VAR) are powerful tools for modeling dynamic interactions among endogenous variables, their standard formulation restricts shock transmission mechanisms and residual covariance structures to remain invariant across time.

The VIX Tupiniquim III addresses precisely this methodological limitation.

In this third and concluding installment of the trilogy, the measurement of Brazilian financial stress is constructed through a TVP-VAR model with Stochastic Volatility (TVP-VAR-SV). The objective is not merely to track whether the financial system has become more volatile, but to model how the cross-market transmission structure itself evolves dynamically.

The proposed indicator synthesizes four core financial dimensions — sovereign risk, monetary policy, equity markets, and bank lending conditions — into a single dynamic barometer.

The label *VIX Tupiniquim* is maintained as a conceptual analogy: it operates as a synthetic gauge of domestic macro-financial stress rather than an exact replica of the CBOE VIX or an options-implied volatility measure[1].

2 Why Use a TVP-VAR for Financial Stress Measurement?

A standard VAR model estimates linear dynamic relationships under the assumption of time-invariant parameters. Economically, this requires assuming that structural linkages across markets remain fixed, regardless of the macroeconomic regime.

For long-horizon financial stress analysis in emerging economies, this assumption is overly restrictive.

A TVP-VAR model relaxes these constraints by allowing parameters to evolve continuously over time. Consequently, the contemporaneous and lagged response to an unexpected shock today can differ substantially from the response observed in another period.

Two distinct dynamic dimensions are captured in this framework:

1. **Time-Varying Parameters:** Autoregressive coefficients and contemporaneous interaction terms vary across the sample, capturing structural shifts in shock transmission;

2. **Stochastic Volatility:** The variances of structural shocks are time-dependent, isolating shifts in the size of exogenous disturbances from changes in transmission mechanics.

Disentangling these two components is essential: a surge in observed market volatility does not necessarily imply that structural transmission has altered. One episode may simply reflect unusually large shocks hitting stable transmission channels, whereas another may stem from fundamentally altered feedback loops between markets.

[1]For earlier installments of the project, see VIX Tupiniquim I and VIX Tupiniquim II.

In economic terms, the TVP-VAR-SV framework enables us to evaluate not only *how volatile the system is*, but also *how shocks propagate across financial channels at each specific point in time*. This constitutes the primary methodological advance of VIX Tupiniquim III.

3 Latent States: What the Model Does Not Directly Observe

The concept of *latent states* reflects a straightforward economic intuition.

Observable market data provide historical trajectories for asset prices, benchmark rates, equity returns, and lending spreads. What they do not directly supply is an explicit, observable time series quantifying, for example, the monthly degree of contagion between sovereign risk and retail credit conditions. This underlying interaction intensity must be inferred.

Time-varying parameters are therefore treated as latent state variables: unobserved structural features of the economic system whose time paths must be filtered from the joint behavior of observable series.

Intuitively, the financial architecture occupies a different state in each period. During periods of stability, sovereign risk spikes may produce negligible spillovers. During severe crises, the same impulse may trigger severe domestic credit and equity contractions.

The model reconstructs these unobserved states statistically. Rather than estimating a single static transmission coefficient for the full sample, the TVP-VAR estimates a continuous trajectory:

$$a_{ij,1}, a_{ij,2}, \ldots, a_{ij,T} \tag{1}$$

Each point along this trajectory represents the estimated state of parameter a_{ij} at period t.

The same formulation applies to stochastic volatility. The time-varying standard deviations of structural shocks are estimated as latent state trajectories:

$$\sigma_{i,1}, \sigma_{i,2}, \ldots, \sigma_{i,T} \tag{2}$$

Thus, "latent" does not denote an arbitrary construct, but an unobservable structural property inferred from data. This distinction is central to the TVP-VAR-SV methodology: it cleanly separates shock size from propagation dynamics.

4 Bayesian Estimation and State Filtering

The presence of time-varying parameters renders standard estimation techniques inapplicable. Unobserved states must be estimated jointly with the remaining hyperparameters of the system.

To address this structure, the model is estimated using Bayesian Markov Chain Monte Carlo (MCMC) methods based on Gibbs Sampling, combined with Kalman state filtering and backward smoothing algorithms[2].

The estimation routine operates across two integrated stages:

- **The Kalman Filter and Smoother** track the dynamic evolution of latent state variables over time, updating the state distributions conditionally on observed data at each step;

[2] The Bayesian state-space estimation methodology with time-varying parameters and stochastic volatility builds on seminal contributions by Carter and Kohn (1994), Kim, Shephard, and Chib (1998), Primiceri (2005), and Nakajima (2011).

- **Gibbs Sampling** updates blocks of parameters iteratively, drawing conditionally from their full posterior distributions. Each completed iteration yields a realization of the joint posterior parameter space.

Intuitively: the Kalman filter reconstructs the historical path of latent states, while Gibbs Sampling explores the distribution of these trajectories to quantify estimation uncertainty.

The algorithm executes 3,000 MCMC draws, discarding the first 1,000 iterations as *burn-in*. The remaining 2,000 draws are retained for posterior inference.

This framework generates full posterior distributions for each parameter at each point in time, enabling the construction of the 16%–84% Bayesian credible bands displayed throughout the empirical figures.

5 Data and System Specification

The empirical dataset comprises 181 monthly observations spanning key segments of the Brazilian financial market.

The system includes four core endogenous variables:

1. **Brazil CDS 5Y ($y_{1,t}$):** Benchmark proxy for sovereign credit risk and global country-risk perception;

2. **Selic Over Rate ($y_{2,t}$):** Annualized policy rate, capturing domestic monetary policy stance;

3. **Ibovespa Return ($y_{3,t}$):** Monthly percentage return of the main Brazilian equity index;

4. **Retail Credit Spread ($y_{4,t}$):** Average interest rate spread on non-earmarked credit operations for individuals, measuring banking financing conditions.

Restricting the system to four core variables preserves broad macroeconomic coverage while ensuring computational stability and MCMC chain convergence[3].

The system with lag order $p = 1$ is expressed as:

$$y_t = c_t + B_{1,t}y_{t-1} + A_t^{-1}\Sigma_t\varepsilon_t \tag{3}$$

where the vector of intercepts c_t, the autoregressive matrix $B_{1,t}$, the contemporaneous interaction matrix A_t, and the diagonal matrix of structural standard deviations Σ_t are all time-varying.

6 Structural Identification and Index Construction

Structural shocks are identified via a lower-triangular Cholesky decomposition of the contemporaneous covariance matrix.

The mapping between reduced-form residuals ϵ_t and orthogonal structural shocks u_t is given by:

$$\epsilon_t = A_t^{-1}\Sigma_t u_t \tag{4}$$

[3]As emphasized by Primiceri (2005), parametric parsimony in TVP-VAR systems is essential to ensure numerical stability and satisfactory mixing properties in Gibbs samplers.

with:

$$A_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a_{21,t} & 1 & 0 & 0 \\ a_{31,t} & a_{32,t} & 1 & 0 \\ a_{41,t} & a_{42,t} & a_{43,t} & 1 \end{bmatrix}, \quad \Sigma_t = \begin{bmatrix} \sigma_{1,t} & 0 & 0 & 0 \\ 0 & \sigma_{2,t} & 0 & 0 \\ 0 & 0 & \sigma_{3,t} & 0 \\ 0 & 0 & 0 & \sigma_{4,t} \end{bmatrix} \tag{5}$$

The ordering reflects an economic hypothesis regarding relative adjustment speeds: sovereign risk (CDS 5Y) responds first, followed by the monetary policy rate (Selic), equity re-pricing (Ibovespa), and finally banking credit spreads:

$$Y_t = \begin{bmatrix} \text{CDS 5Y}_t \\ \text{Selic Over}_t \\ \text{Ibovespa Return}_t \\ \text{Retail Spread}_t \end{bmatrix} \tag{6}$$

The time-varying reduced-form conditional covariance matrix is constructed as:

$$\Omega_t = A_t^{-1}\Sigma_t\Sigma_t'(A_t^{-1})' \tag{7}$$

The system-wide stress scale is defined by the square root of the maximum eigenvalue of Ω_t:

$$s_t = \sqrt{\lambda_{\max}(\Omega_t)} \tag{8}$$

The resulting index is normalized to have a historical sample mean of 100:

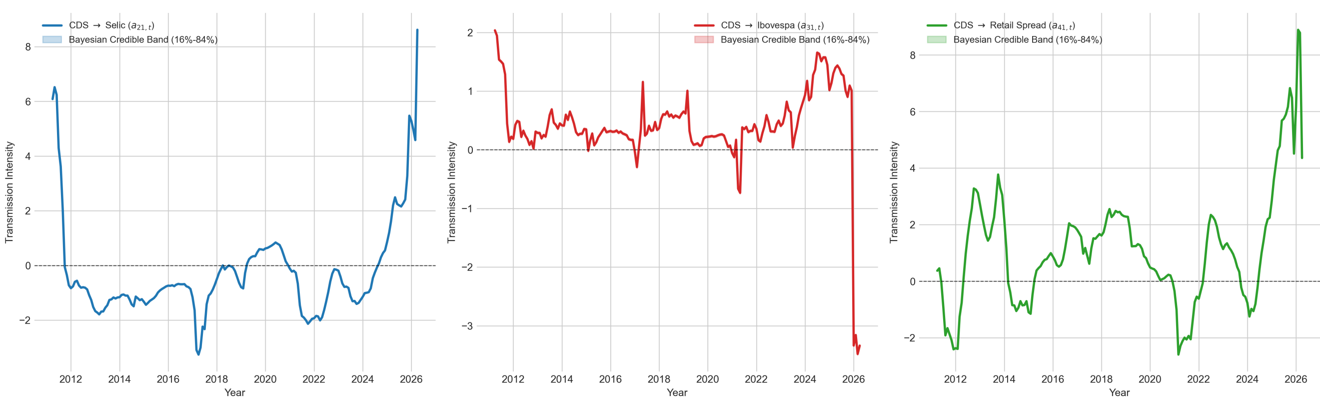
$$\text{VIX Tupiniquim III}_t = 100 \times \left(\frac{s_t}{\frac{1}{T}\sum_{t=1}^{T} s_t} \right) \tag{9}$$

Values above 100 indicate periods of above-average domestic financial stress.

7 Results: Dynamic Shock Transmission

Figure 1 displays the evolution of contemporaneous transmission coefficients from matrix A_t^{-1}, measuring the direct impact of sovereign risk (CDS 5Y) shocks on remaining domestic markets.

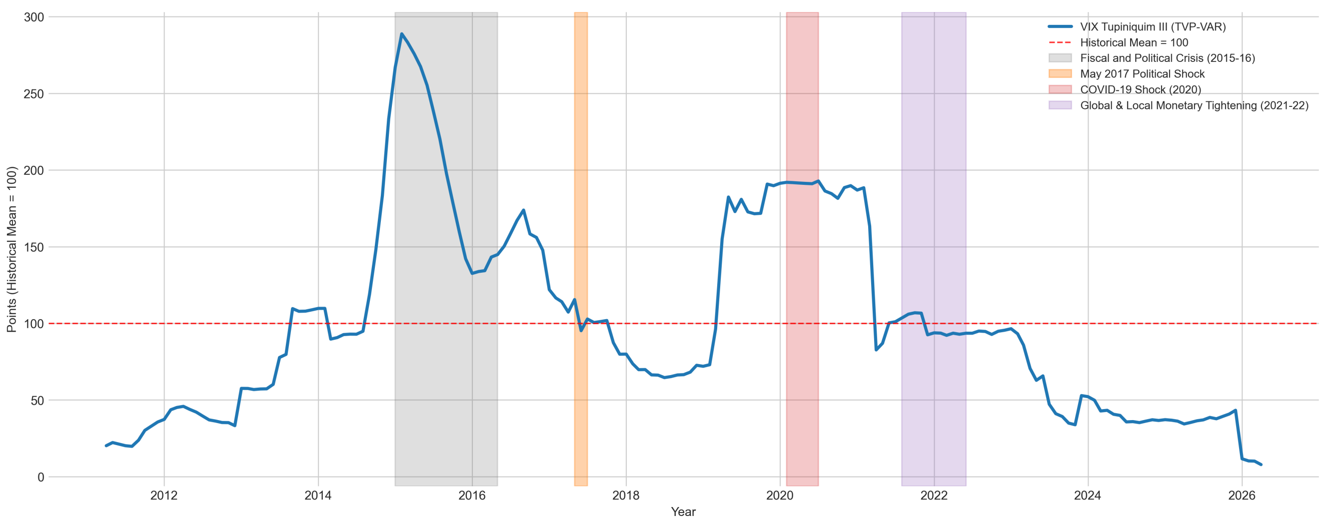
Figure 1: **Contemporaneous Transmission Coefficients from CDS 5Y (A_t^{-1})**



Parameter trajectories demonstrate that shock transmission is non-constant over time. The contemporaneous link between sovereign risk and Selic ($a_{21,t}$) remains negative across extended periods, with noticeable upward inflections in 2011 and 2025–2026 driven by global monetary tightening and domestic risk-premium repricing. In the equity channel ($a_{31,t}$), transmission remains relatively stable across normal regimes but exhibits severe negative spikes during acute distress. Transmission to retail credit spreads ($a_{41,t}$) displays pronounced regime switches, rising sharply toward the end of the sample.

Figure 2 plots the VIX Tupiniquim III alongside key macro-financial stress episodes in modern Brazilian economic history.

Figure 2: **VIX Tupiniquim III: Historical Stress Events and Tail Regimes**

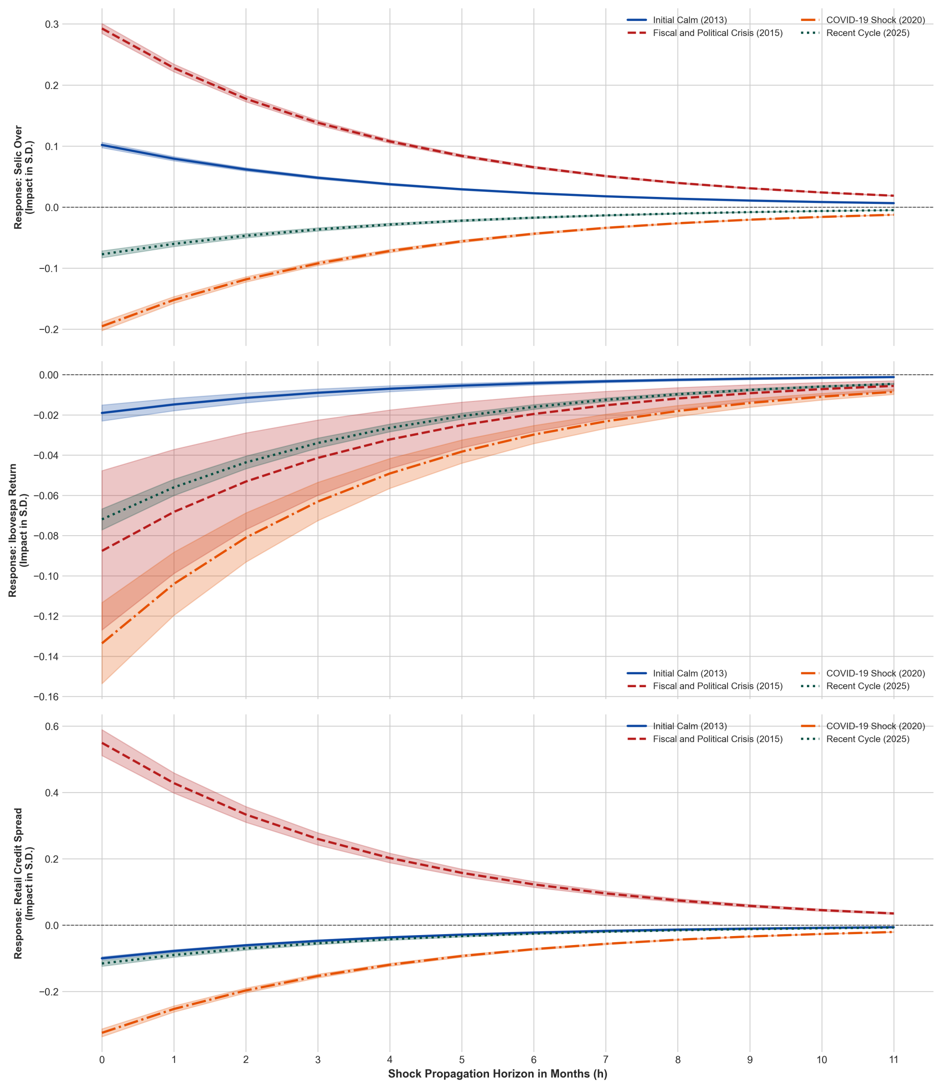


The indicator reveals contrasting temporal profiles across crises. The 2014–2016 downturn exhibits a cumulative, protracted deterioration, reaching a historical peak near 290 points before receding. In contrast, the 2020 COVID-19 shock is characterized by an abrupt jump to 190 points, sustaining an elevated plateau for nearly two years despite the policy rate reaching a historical low of 2.0%. From 2023 onward, the index settles into subdued levels below its historical average.

8 Time-Varying Impulse-Response Functions

A key advantage of the TVP-VAR framework is evaluating Time-Varying Impulse-Response Functions (TV-IRF). Figure 3 compares responses to a one-standard-deviation CDS shock across four representative regimes: 2013 (Initial Calm), 2015 (Fiscal and Political Crisis), 2020 (COVID-19 Shock), and 2025 (Recent Cycle).

Figure 3: **Time-Varying Impulse-Response Functions (TV-IRF) by Regime**

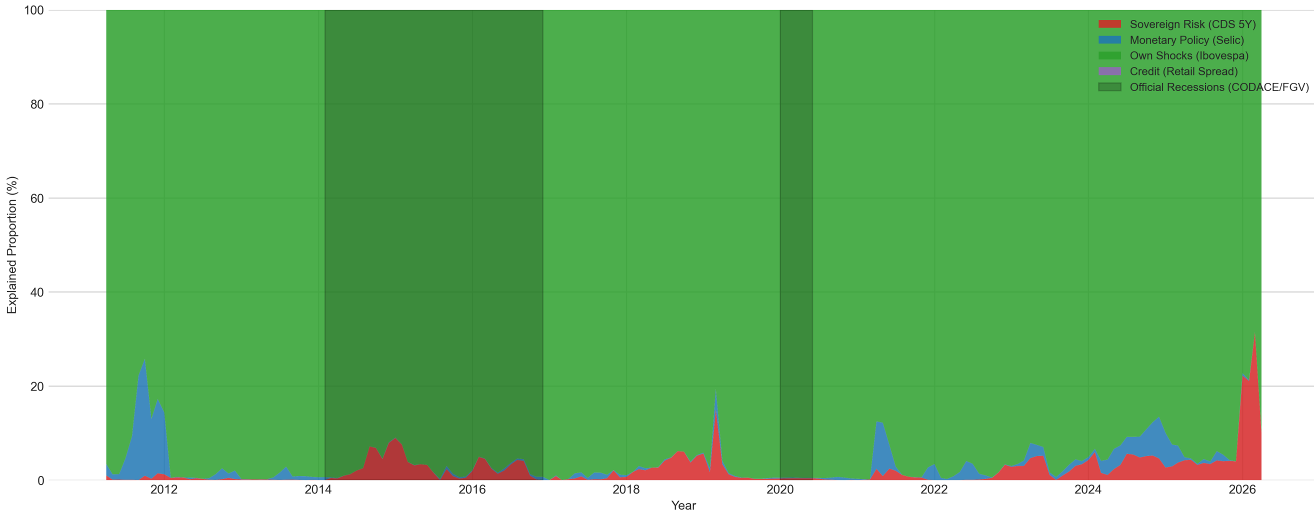


The empirical responses show that shock propagation varies substantially in magnitude and direction across regimes. Monetary policy responses display noticeable asymmetry: in the 2015 regime, the contemporaneous response reaches +0.30 standard deviations and remains positive until $h = 11$, consistent with a defensive monetary stance amid deteriorating fiscal conditions. In equities, Ibovespa experiences an immediate contraction of -0.135 standard deviations at $h = 0$, followed by rapid absorption between $h = 6$ and $h = 8$. Retail credit spreads show pronounced asymmetry, shifting from a strong positive widening in 2015 (+0.55 standard deviations) to a contemporaneous contraction in 2020 (-0.30 standard deviations).

9 Variance Decomposition

Figure 4 presents the dynamic Forecast Error Variance Decomposition (TV-FEVD) for Ibovespa monthly returns over a 12-month horizon.

Figure 4: **Dynamic Forecast Error Variance Decomposition (TV-FEVD) for Ibovespa Return (12 Months)**

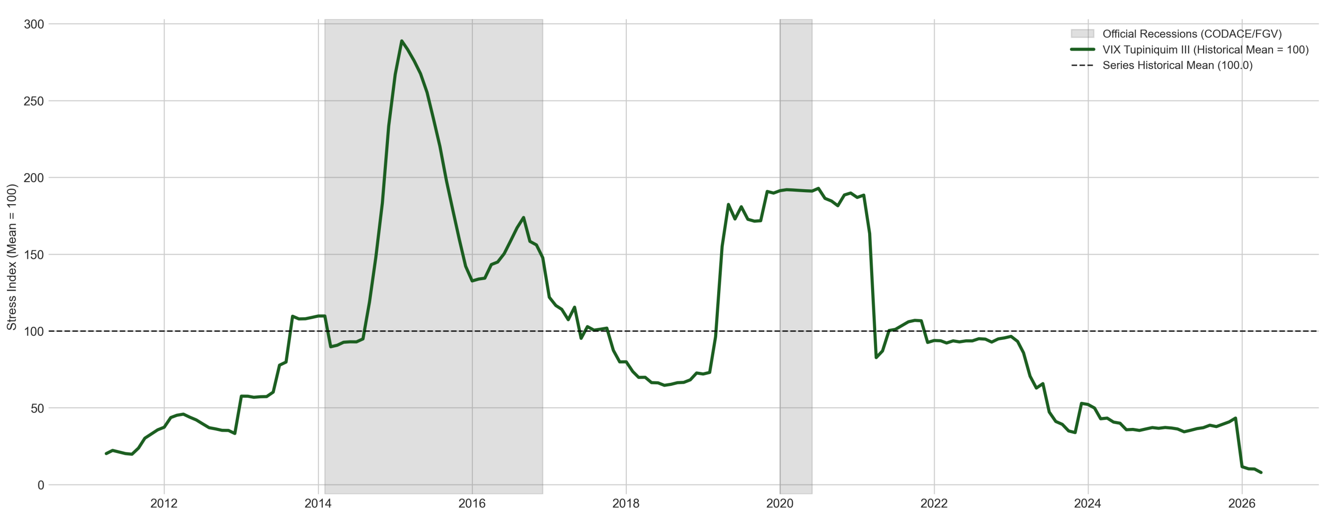


Equity returns are predominantly driven by own structural shocks, accounting for 70% to 90% of total variance across the sample. The remaining variance composition shifts across regimes: the policy rate operates as an intermediate channel during specific cycles, while credit spreads exert a marginal influence. The contribution of sovereign risk (CDS 5Y) shocks rises noticeably during major stress episodes and escalates sharply starting in late 2025, a pattern temporally consistent with fiscal debates surrounding the trajectory of General Government Gross Debt (GGGD).

10 Historical Trajectory and Real Economic Cycles

Figure 5 compares the VIX Tupiniquim III against official recession periods dated by CODACE/FGV.

Figure 5: **VIX Tupiniquim III and Official CODACE/FGV Recessions**



The comparison confirms that financial stress and real business cycle contractions operate under distinct timelines:

- **2014–2016 (Protracted Recession):** Financial stress closely aligns with the real downturn, rising steadily through the fiscal and political crisis to its historical apex.
- **2020–2021 (Pandemic Shock and Persistence):** While the official CODACE recession was confined to two quarters in early 2020, financial stress remained elevated over a considerably longer window. The indicator reflects joint market volatility and risk pricing, which do not automatically dissipate with the statistical end of GDP contraction.
- **2023–2026 (Transition and Normalization):** The indicator transitions into a regime persistently below its 100 baseline, signaling moderate aggregate financial volatility.

11 VIX Tupiniquim III and Brazil EPU: Is There Temporal Precedence?

To evaluate whether market-based financial stress leads news-based economic uncertainty, the VIX Tupiniquim III is tested against the Economic Policy Uncertainty Index for Brazil (Brazil EPU)[4].

ADF unit root tests indicate that VIX Tupiniquim III is non-stationary in levels ($p = 0.2341$) and stationary in first differences ($p < 0.001$), while Brazil EPU is stationary in levels ($p = 0.0027$).

Table 1: **Temporal Precedence Tests: VIX Tupiniquim III and Brazil EPU**

Tested Direction (H_0)	**Lag**	**F-Statistic**	**p-Value**	**Decision at 5%**
ΔVIX III $\rightarrow$ EPU	1	0.0771	0.7811	Do not reject H_0
ΔVIX III $\rightarrow$ EPU	2	1.2631	0.2857	Do not reject H_0
ΔVIX III $\rightarrow$ EPU	3	0.7818	0.5062	Do not reject H_0
EPU $\rightarrow$ ΔVIX III	1	1.0432	0.3086	Do not reject H_0
EPU $\rightarrow$ ΔVIX III	2	0.5098	0.6018	Do not reject H_0
EPU $\rightarrow$ ΔVIX III	3	0.7398	0.5297	Do not reject H_0

Note: The Toda-Yamamoto procedure ($k = 2$, $d_{\max} = 1$) also fails to reject the null hypothesis of no temporal precedence in levels: VIX III $\rightarrow$ EPU ($p = 0.3287$) and EPU $\rightarrow$ VIX III ($p = 0.3312$).

The absence of linear precedence confirms that the two metrics capture complementary dimensions of macroeconomic uncertainty: one grounded in pricing decisions across financial markets and the other anchored in media and policy discourse.

[4]The Brazil EPU index follows the methodology of Baker, Bloom, and Davis (2016). Robust level causality testing follows Toda and Yamamoto (1995).

12 Concluding Remarks

The VIX Tupiniquim III concludes the trilogy by addressing the question: *how does shock transmission across financial markets change over time?* While VIX I condensed linear relationships via PCA and VIX II captured non-linear threshold effects via XGBoost/SHAP, VIX III models the dynamic evolution of both structural parameters and stochastic volatilities.

The results confirm that Brazilian macro-financial dynamics are state-dependent. Market responses to shocks depend crucially on prevailing regime conditions. The VIX Tupiniquim III constitutes a relative gauge of domestic financial stress, tracking both system-wide stress accumulation during crises and subsequent return to normalcy.

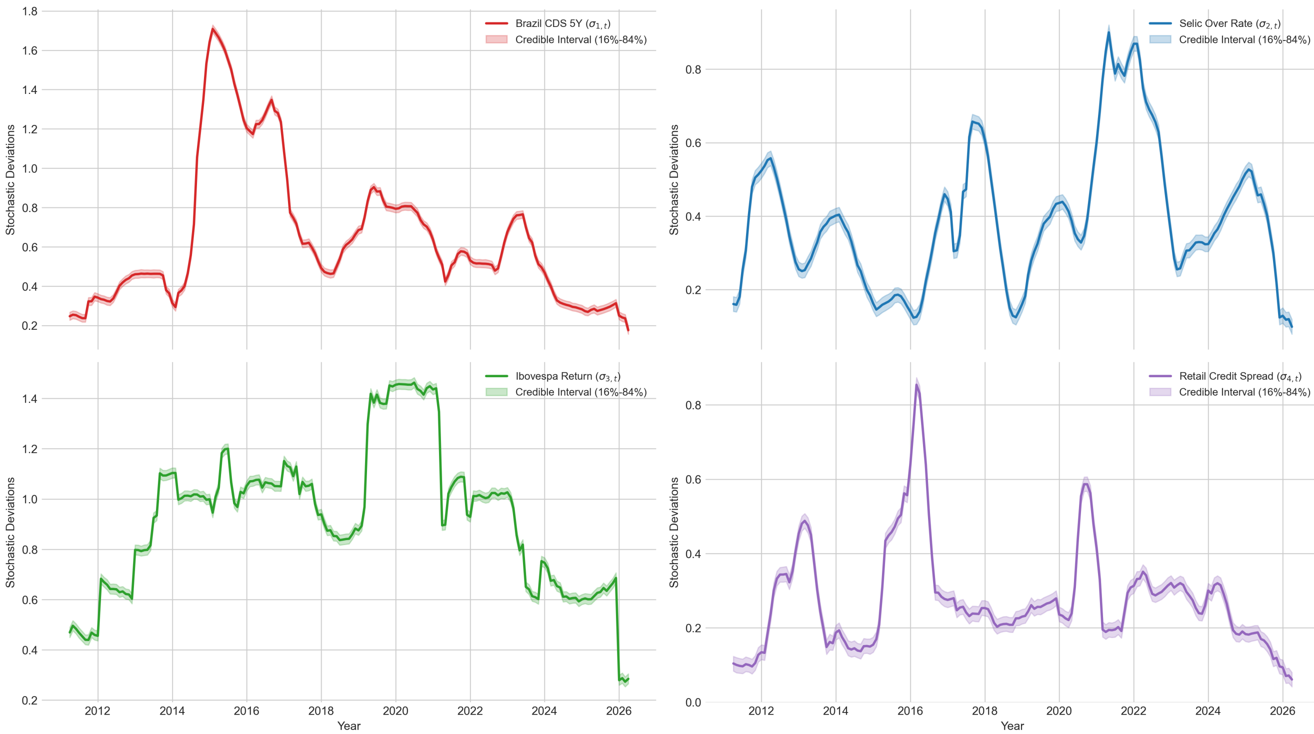
Finally, the **VIX Tupiniquim Trilogy** does not aim to designate a single "optimal model", but rather to demonstrate how distinct econometric and machine learning methodologies can be combined to enrich macro-financial monitoring and quantitative decision-making.

Methodological Appendix and Supplementary Charts

A. Stochastic Volatility Latent States

Figure A1 presents the estimated trajectories of stochastic log-volatilities ($\sigma_{i,t}$) for each market in the system. The estimates confirm distinct crisis signatures: the 2015–2016 downturn exhibited persistent volatility in CDS 5Y and retail credit spreads, whereas the 2020 COVID-19 shock produced acute, transitory volatility spikes in the policy rate, equity returns, and sovereign risk.

Figure A1: **Trajectories of Stochastic Volatility Latent States (Σ_t)**



B. Cross-Market Contemporaneous Transmission Matrix

The contemporaneous mapping between reduced-form residuals ϵ_t and orthonormal structural shocks u_t is defined by the lower-triangular system:

$$\begin{bmatrix} \epsilon_t^{\text{CDS}} \\ \epsilon_t^{\text{Selic}} \\ \epsilon_t^{\text{Ibov}} \\ \epsilon_t^{\text{Spread}} \end{bmatrix} = A_t^{-1}\Sigma_t u_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a'_{21,t} & 1 & 0 & 0 \\ a'_{31,t} & a'_{32,t} & 1 & 0 \\ a'_{41,t} & a'_{42,t} & a'_{43,t} & 1 \end{bmatrix} \begin{bmatrix} \sigma_{1,t} & 0 & 0 & 0 \\ 0 & \sigma_{2,t} & 0 & 0 \\ 0 & 0 & \sigma_{3,t} & 0 \\ 0 & 0 & 0 & \sigma_{4,t} \end{bmatrix} \begin{bmatrix} u_t^{\text{CDS}} \\ u_t^{\text{Selic}} \\ u_t^{\text{Ibov}} \\ u_t^{\text{Spread}} \end{bmatrix} \tag{10}$$

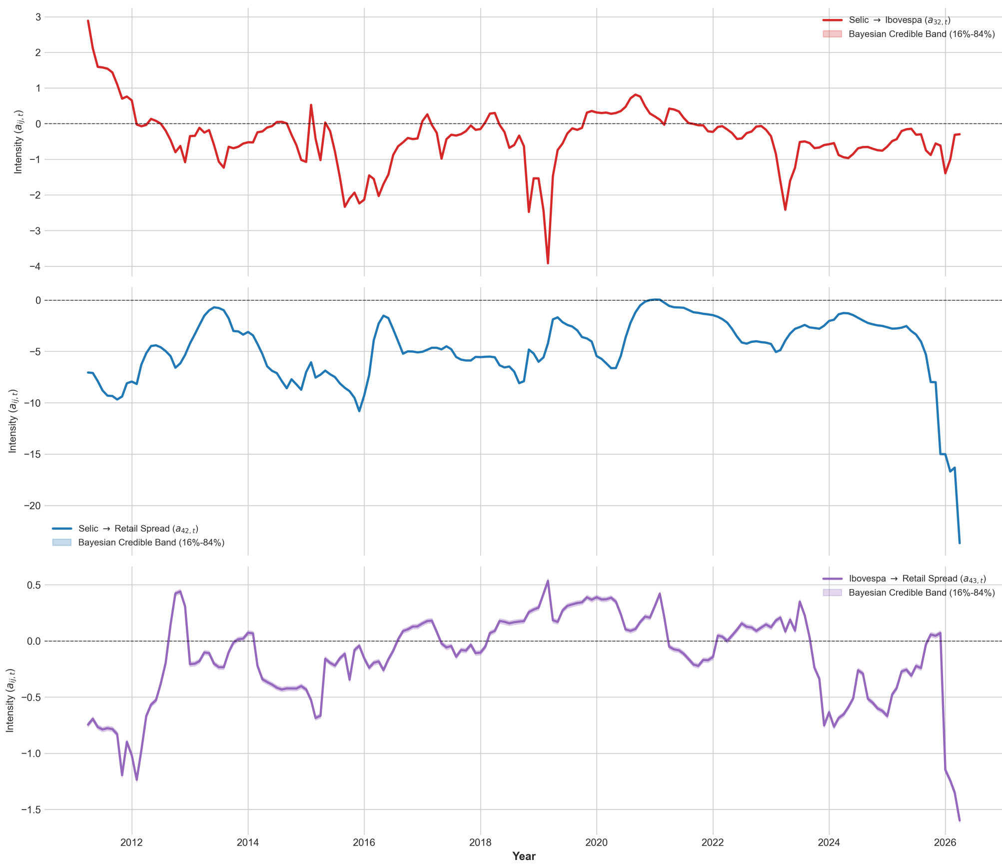
where ϵ_t denotes reduced-form innovations, A_t^{-1} maps contemporaneous cross-market transmission, and $u_t \sim \mathcal{N}(0, I)$ represents structural shocks. Economically:

- **CDS 5Y:** Responds contemporaneously only to its own structural shock (1);
- **Selic Over:** Responds to own shocks (1) and sovereign risk spillovers ($a_{21,t}$);
- **Ibovespa:** Responds to own shocks (1), sovereign risk ($a_{31,t}$), and policy rates ($a_{32,t}$);

- **Retail Spread:** Responds to own shocks (1), sovereign risk ($a_{41,t}$), policy rates ($a_{42,t}$), and equities ($a_{43,t}$).

Figure A2 illustrates contemporaneous interaction coefficients across the remaining domestic market pairs.

Figure A2: **Contemporaneous Cross-Market Transmission Matrix** (A_t^{-1})



C. Comparative Structural Regimes Summary

Table A1 summarizes the estimated TVP-VAR parameter configurations across selected historical regimes.

Table A1: **Structural TVP-VAR Configuration across Historical Regimes**

Historical Regime	**VIX III**	**CDS → Selic** (a_{21})	**CDS → Ibov** (a_{31})	**CDS → Spread** (a_{41})	**FEVD CDS → Ibov (12m)**
Initial Calm (2013)	41.2	Negative (Low)	Positive (Moderate)	Negative	≈ 1.5%
Fiscal and Political Crisis (2015)	289.4	Negative	Positive (Stable)	Positive (Strong)	≈ 8.5%
COVID-19 Shock (2020)	192.1	Positive (Transitory)	Negative Inflection	Negative	≈ 14.0%
Monetary Tightening (2021–22)	106.8	Negative	Positive	Negative	≈ 4.2%
Recent Cycle (2025)	39.5	Upward Inflection	Negative Inversion	Positive (Pronounced)	≈ 7.8%
Sample End (2026)	12.4	Sharp Rise	Extreme Negative	Sharp Rise	> 25.0%